

# Harshwardhan Patil

📍 Arlington, TX

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## Summary

Full-Stack Engineer with **3+ years of experience** building scalable, distributed systems and high-performance APIs across frontend, backend, and cloud. Skilled in **system design**, CI/CD on AWS, production monitoring, and delivering **fault-tolerant, data-consistent services** at scale in **Agile** environments.

## Technical Skills

- **Languages & Frameworks:** Java, TypeScript, JavaScript, SQL, Spring Boot, Angular, Node.js, Python, FastAPI
- **Cloud & DevOps:** AWS (EC2, ECS, S3, RDS, CloudFront, ALB, Secrets Manager, Lambda), Docker, Jenkins, Git, GitHub Actions, Nginx, Cloudflare
- **Databases:** PostgreSQL, Redis, MySQL, Neo4j, Elasticsearch, MongoDB, Schema design, Database indexing, Query optimization, Caching strategies
- **Testing & Security:** JUnit, Mockito, Jasmine, Vitest, ESLint, SonarQube, Postman, JWT, RBAC, HTTPS/TLS, CORS, CSRF/XSS protection
- **Design & Architecture:** System Design, Design Patterns, SOLID principles, Distributed backend services, Concurrency handling, Session management
- **AI & ML:** Spring AI, OpenAI API, Prompt Engineering, Embeddings, Vector Search, RAG Pipelines, pgvector, ChromaDB, Tool Calling, Pydantic

## Professional Experience

**Software Engineer, Clarivate Analytics** – Bangalore, India

Aug 2023 - Jul 2024

- Developed scalable **Spring Boot** REST APIs with DTO contracts and standardized error handling, reducing breaking changes by **30%** across releases.
- Implemented **JWT**-based multi-session management across backend services, capping concurrent logins at 12 per user with token invalidation on breach.
- Optimized **PostgreSQL** schemas with **composite indexing** across reporting and search workloads for drug and patent datasets, cutting full table scans.
- Containerized and deployed microservices to **EC2** using **Docker** and **Jenkins CI/CD** pipelines, supporting **50+ production releases** with zero rollbacks.
- Enforced **role-based access control** across backend and frontend layers, applying tiered and region-based data visibility for **10+ access variants**.
- Implemented low-latency **Elasticsearch**-powered typeahead and autocomplete, improving search response times by **40%** over large-scale datasets.
- Developed dynamic PPT and PDF export pipelines using **PptxGenJS** and **jsPDF**, generating **100+ customized reports** with user-selected filters.
- Optimized **PostgreSQL** query logic for PPT export, decoupling bulk multi-filter chart fetching from single-chart UI rendering to reduce DB round-trips.
- Achieved **90%** test coverage using **JUnit** and **Mockito** for backend and **Jasmine** for Angular, covering unit, branch, and line coverage across all modules.

**Associate Software Engineer, Clarivate Analytics** – Bangalore, India

Aug 2021 - Jul 2023

- Integrated **Elasticsearch** into **Spring Boot**, enabling advanced full-text queries across **millions of drug records** and reducing search latency by **30%**.
- Migrated **5+ core business flows** from Angular 8 to Angular 13, rebuilding components with a modular architecture and improved state management.
- Built secure request-handling across backend services using **Spring Security**, **CORS/CSRF**, and input validation to guard against unauthorized access.
- Boosted Angular performance by **20–25%** via lazy loading, **RxJS** streams, **OnPush** change detection, **memoization**, and optimized data-fetching patterns.
- Built **D3.js** chart components with grouping and filtering aggregations in Angular, powering **15+ analytics views** across drug intelligence dashboards.
- Architected a configurable **Angular** table framework with sorting, pagination, and filters; adopted across **3 internal teams**, cutting duplication by **50%**.

## Projects

**TinyURL — URL Shortener** 🔄 🔗

Feb 2026 - Mar 2026

Production-grade URL shortening service with QR code generation, link expiry, and a fully automated cloud deployment pipeline.

**Tools:** Spring Boot, Angular, PostgreSQL, AWS (EC2, RDS, S3, CloudFront, ALB), Docker, Cloudflare, Nginx, Lambda, EventBridge, SQS DLQ, CloudWatch

- Designed 6-layer request path across Cloudflare, CloudFront, ALB, Nginx, **Spring Boot**, and **PostgreSQL**, separating SPA delivery from API and redirects.
- Hardened API traffic with **Nginx rate limits**, **Bucket4j token buckets**, **layer-7** bot/scanner filters, per-IP size caps, and strict **HSTS/CSP** secure headers.
- Designed serverless EC2/RDS scheduler with **Lambda**, **EventBridge**, CloudWatch alarms, SNS, and **SQS DLQ**, reducing monthly cloud cost by **~40%**.
- Built zero-SSH **CI/CD** with GitHub Actions, AWS OIDC, SSM RunCommand, Docker smoke tests, cosign signing, and auto CloudFront cache invalidation.

**diagram-sync — Diagram-as-Code CLI** 🔄 🔗 📄 3,796

May 2026 - Present

Published npm package that renders architecture diagrams from 7 source formats into versioned SVG/PNG outputs via `npm diagram-sync`.

**Tools:** TypeScript, Node.js, Vitest, ESLint, GitHub Actions

- Engineered **provider-registry architecture** mapping 14+ file extensions to isolated rendering adapters with runtime graceful degradation on missing tools.
- Developed recursive repo scan, `--files`, and `--changed` via `git diff HEAD` with untracked support, enabling CI to regenerate only PR-changed files.
- Architected **tiered format resolution** across CLI flag, job, and global config with per-provider validation and actionable errors for unsupported formats.

**smart-anytool-agent — Research-Backed Tool-Calling Agent** 🔄 🔗 🔗

Jul 2026 - Aug 2026

Implements **SMART** and **AnyTool** techniques with self-awareness gating, hierarchical tool filtering, and self-reflection retry loops.

**Tools:** Python, OpenAI API, Pydantic, httpx, GitHub Actions

- Implemented **SMART self-awareness** layer gating tool calls via LLM pre-check, cutting unnecessary invocations by **24%** and improving accuracy by **37%**.
- Engineered **hierarchical tool filtering** with LLM category classification and progressive widening fallback, exposing only relevant tool subsets per query.
- Built **self-reflection recovery loop** with parallel async tool execution and **Pydantic**-validated dispatch, handling multi-step agentic queries end-to-end.

**RAG Document Engine** 🔄

Jun 2026 - Present

End-to-end RAG system with multi-format ingestion, hybrid retrieval, LLM answer generation, and role-based document access control.

**Tools:** Python, NumPy, OpenAI API, ChromaDB, sentence-transformers

- Built multi-format ingestion (PDF, Word, Markdown) with overlapping chunking, deduplication, and **1536-dim** L2-normalized embeddings into **ChromaDB**.
- Engineered **hybrid retrieval** via ensemble sparse **BM25** + cross-encoder reranking with **tiktoken** token budgeting for citation-grounded LLM generation.
- Implemented multi-turn chat with history truncation and **RBAC-scoped metadata filtering** in **ChromaDB** enforcing per-user document access at retrieval.

## Education

University of Texas at Arlington

Master's in Software Engineering

GPA: 3.6 / 4.0

Aug 2024 - May 2026